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SCENT DISPENSERS/ABSORBERS AND SCENT CARTRIDGES THEREFOR

Abstract

A scent dispenser/absorber includes a container, an invertible lid reversibly attached to an open end of the container, a vented cage, and a scent cartridge housed within the cage. In an “off” condition or state, the scent cartridge/cage combination is received in an interior chamber of the container, where the scent cartridge is out of communication with the surrounding environment and therefore incapable of performing scent dispensing function. The scent dispenser/absorber can be switched to an “on” condition or state by manually inverting the lid and, in some cases, the associated cage/scent cartridge combination, such that the scent cartridge can be positioned externally of the container, where it can communicate with the surrounding (i.e., external) environment to perform methods useful for attracting game, calming pets, medical diagnostics, aroma and insomnia therapy, and dieting.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. Non-provisional patent application Ser. No. 18/634,727 filed Apr. 12, 2024, which claims priority to U.S. Provisional Patent Application Ser. No. 63/459,156 filed Apr. 13, 2023 and is a continuation-in-part of U.S. Non-provisional patent application Ser. No. 17/991,612 filed Nov. 21, 2022, which is a continuation-in-part of PCT Patent Application Serial No.

PCT/US2021/033740 filed May 21, 2021, which claims priority to U.S. Provisional Patent Application Ser. No. 63/028,433 filed May 21, 2020, U.S. Provisional Patent Application Ser. No. 63/091,786 filed Oct. 14, 2020, and U.S. Provisional Patent Application Ser. No. 63/132,303 filed Dec. 30, 2020 and incorporates by reference the entire disclosure of each of the aforesaid applications.

FIELD OF THE INVENTION

[0002] The present invention relates to dispensers of scents into the environment and/or absorbers of odors from the environment.

BACKGROUND OF THE INVENTION

[0003] Various forms of dispensers of scents into the environment, such as household or other building interior environments, or devices for removing odors from such environments, are known in the art. Many of these dispensers and/or devices have a multi-component design requiring them to be assembled with the use of independent fasteners and/or tools.

[0004] Devices are known which carry and emit scent (typically as a vapor) to the ambient environment. In some cases, these known devices are in the form of foams, fibers, gels, or porous materials that have been loaded with scents, typically in the form of a liquid or a solid. Such scented devices are often housed in a vessel or receptacle which contains the scent-containing foams, fibers, gels, or porous materials. In use, these known devices are often designed and adapted to deliver wet air to a room or other large space.

SUMMARY OF THE INVENTION

[0005] An object of at least one embodiment of the present invention is to provide a versatile scent-dispensing/absorbing apparatus that is efficient and effective in use, while maintaining simplicity and economics of manufacture and operation.

[0006] Embodiments of the present invention have, as an additional object, the ability to dispense scents via molecules (i.e., “dry air”), free from mists, sprays, aerosols and liquid, while requiring no “active” driving mechanisms such as volatile organic compounds (i.e., VOCs), chemicals,

electricity, heat, or similar propellant enablers to achieve diffusion. The ability to minimize propellants, especially moisture-based ones in accordance with this object, is advantageous in that it may mitigate the spread of contagious disease, such as the one which plagued the world during the recent COVID crisis.

[0007] Another object of embodiments of the present invention is to provide scents dedicated to different functions, such as study aid scents, holiday scents, soothing scents, teleconferencing scents, desk/work scents, wellness related scents, smell retraining therapy as a treatment protocol for anosmia and hyposmia, etc.

[0008] A further object of embodiments of the present invention is to provide for personal, portable scent diffusers that can be used anywhere and at any time.

[0009] It is yet another object of embodiments of the present invention to allow essential oils to be diffused in pure, unadulterated form without artificial propellants. In accordance with this object, embodiments of the present invention accomplish scent dispensing/absorbing functions with ambient air used as the propellant.

[0010] Another object of at least one embodiment of the present invention is to provide and enable the transfer of oil-based scents, while minimizing or eliminating a user's interaction with any liquid-state scent material.

[0011] Yet another object of at least some of the embodiments of the present invention is to ensure that the odorant scents diffused from a device are the same as those initially infused therein, thereby minimizing adulteration of the diffused scent.

[0012] It is still a further object of various embodiments of the present invention to operate free of liquids, with such embodiments utilizing scent cartridges containing a minimal amount of liquid odorants. In accordance with this object, embodiments of the present invention operate via a diffusion process free of liquid droplets or aerosols.

[0013] It is yet a further object of embodiments of the present invention to provide a device that can be easily and manually switched or toggled between “on” and “off” configurations, while also allowing for the controlled direct transmission of scent (i.e., varying the rate of diffusion/absorption).

[0014] An additional object of various embodiments of the present invention is to provide scent-absorbing or diffusing functions lasting multiple months in typical use.

[0015] It is yet another object of at least one embodiment of the present invention to provide a device that is compatible with any odorant used now or in the future.

[0016] A still further object of various embodiments of the present invention is to provide a device designed such that essential oils are dispensed in a way that avoids damage to finished surfaces or fabrics.

[0017] An additional object of various embodiments of the present invention is to allow for personal and portable scent diffusion, whereby scent can be delivered, without batteries or plug-in electricity, directly to the personal space of an individual, such as while the individual is driving a vehicle or is seated at a desk, rather than to an entire room or similar large space.

[0018] A not necessarily final object of at least one embodiment of the present invention is to provide a scent insert adapted for use with a scent dispenser and configured to control: 1) the length of effective time of scent diffusion; 2) the intensity of the diffused scent; and 3) the exponential decay curve of the diffused scent (e.g., promoting generally longer, linear curves).

[0019] In accordance with the aforementioned objects, an embodiment of the present invention relates to a novel “dry air” scent dispenser, which includes a cylindrical container, a reversible lid, a vented cage and a scent cartridge. The scent cartridge fits inside the cage, which can then be attached to one side of the lid. The lid is configured to be received by the container in two different conditions (i.e., an “off” condition and an “on” condition). In the “off” condition, the cage and scent cartridge are confined inside the container, where they do not interface with the outside air to absorb/dispense scents. To switch to the “on” condition, a user removes the lid, inverts it such that

the scent cartridge and cage are outside the container and then places the lid back on the container in an inverted position. In such an “on” condition or inverted position, scents are dispensed to/absorbed from the surrounding air. When not in use, the lid can be inverted once more by a user and replaced on the container to return the device to its original position and therefore its “off” condition.

[0020] In one particular embodiment, the aforementioned insert is placed in (i.e., encased or otherwise enclosed within) a fabric form, examples of which include, but are not limited to, a pillow, cushion, pouch, sachet or similar enclosure. In use, scent molecules originating from the odorant are passively and continuously released from the insert's reticulated, hydrophilic inner core. The scent molecules released from the inner core of the insert pass through the insert's thermoplastic outer skin and then the fabric form, before being diffused into the surrounding environment to provide a faithful iteration of the original odorant or scent for sampling, fragrancing or other purposes, such as aroma therapy and insomnia therapy. In such an embodiment of the present invention, the thermoplastic membrane functions as a barrier between the fabric form and the insert's scent-laden inner core, thereby inhibiting wicking of scent from the inner core to the fabric form. Scent residue on the fabric form due to such wicking is undesirable because it could result in increased diffusion rates and unpleasant feel, while potentially damaging foreign objects coming into contact with such residue. The same barrier functionality is realized in accordance with an alternate embodiment of the present invention in which the thermoplastic membrane is incorporated into the fabric form before being combined with the reticulated, hydrophilic composite material of the inner core.

[0021] Another embodiment of the present invention involves an apparatus for dispensing or absorbing scent, comprising a housing having an open end, a closed end opposite the open end, and an interior chamber between the open and closed ends. The apparatus also includes an invertible lid having an associated compartment configured to allow the egress or ingress of scent therefrom or thereto. The lid is attachable to the open end of the housing in a first orientation, in which the lid's compartment is positioned entirely within the interior chamber of the housing between the lid and the closed end of the housing, and in a second orientation, in which the lid's compartment is positioned externally of the housing.

[0022] In accordance with another particular embodiment of the present invention, a scented material is in the form of a combination, which comprises a reticulated hydrophilic composite material; an optional odorant contained within the composite material; and a thermoplastic membrane surrounding or enveloping the composite material. The reticulated hydrophilic composite material may be in the form of a subcombination, which comprises a hydrophobic scaffold made from a hydrophobic material, and a hydrophilic material which coats the hydrophobic material.

[0023] A still further embodiment of the present invention involves another combination, namely, the aforementioned scented material and a fabric enclosure that envelops the scented material. The fabric enclosure can form part of a useful article (e.g., a pillow, cushion or the like)

[0024] The present invention also involves scent dispensing methods, a number of which include the step of providing an apparatus which includes: a housing having an open end, a closed end opposite the open end, and an interior chamber between the open and closed ends; an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, the lid being attachable to the open end of the housing in a first orientation, in which the lid's compartment is positioned entirely within the interior chamber of the housing between the lid and the closed end of the housing, and in a second orientation, in which the lid's compartment is positioned externally of the housing. These methods further include the steps of providing the lid's compartment with a scented material and selecting a scent adapted to perform any one or more of a variety of functions, such as attracting game, calming a pet animal, aiding medical diagnostics of a patient, facilitating aroma therapy of an individual (e.g., smell retraining therapy), facilitating the dieting of an

individual, and facilitating insomnia therapy of an individual.

[0025] Another embodiment of the present invention involves a system that includes a collection of interchangeable scent cartridges adapted for use in combination with at least one type of compatible scent dispenser, which could be configured to simultaneously receive a plurality of scent cartridges having the same or different scents. When two or more scent cartridges are combined for use in a single dispenser and each cartridge is provided with a scent different from the others, a user of the dispenser can design customized composite (i.e., hybrid) scents.

[0026] The system also includes a repository for the collection of interchangeable scent cartridges, as well as a subscription service for users of at least one type of compatible scent dispenser, the subscription service allowing users to place orders for one or more of the interchangeable scent cartridges, which can be implemented as lightweight and compact discs. Due to their lightweight and compact design, the scent cartridges or discs can be removed at will from their associated dispensers, stored in small foil lined bags (such as those they are shipped in), and reused at will. In this way, a new scent cartridge, or the same scent cartridge, can be inserted or reinserted into the same or a different dispenser, thereby providing a user with the flexibility to indulge a variety of different and diverse scent-sampling experiences.

[0027] An object of another embodiment of the present invention is to provide a versatile scent-dispensing/absorbing apparatus that is efficient and effective in use, while maintaining simplicity and economics of manufacture and operation. This object, when achieved in accordance with the present invention, results in a commercially viable product.

[0028] The present invention has, as an additional object, the ability to dispense scents via molecules (i.e., “dry air”), rather than via droplets, moisture, etc., while requiring no “active” driving mechanisms such as volatile organic compounds (i.e., VOCs), chemicals, electricity, heat or similar propellant enablers to achieve diffusion. In view of recent public healthcare crises, minimizing propellants, especially those moisture-based ones, may help to mitigate the spread of contagious disease.

[0029] A further object of the present invention is to provide for personal, portable scent diffusers that can be used anywhere and at any time. For example, the present invention is manifested by a device that can be easily and manually switched between “on” and “off” configurations.

[0030] In accordance with the aforementioned objects, the present invention relates to a novel “dry air” scent dispenser, which includes a disc-shaped base, a detachable and invertible lid, a vented cage, and a scent cartridge housed in a compartment formed in the base. The lid can be interchangeably screwed on or otherwise attached to an open end or a closed end of the base, depending upon whether the scent dispenser is in its “off” condition or “on” condition. In the “off” condition, the lid is applied to the open end of the base such that it covers the vented cage to thereby prevent the scent cartridge from interfacing with the outside air (i.e., surrounding atmosphere) to absorb/dispense scents. To switch to the “on” condition, a user removes the lid (e.g., by unscrewing it), whereby the scent cartridge and cage are exposed to the surrounding atmosphere. The lid can then be screwed onto the closed end of the base in an inverted position for storage and potential attachment to a foreign object (e.g., a sun visor for a motor vehicle). In such an “on” condition, scents are dispensed to/absorbed from the surrounding atmosphere. When not in use, the lid can be removed from the closed end of the base and reattached to the base's open end, thereby covering the cage and scent cartridge so as to return the scent dispenser to its “off” condition.

[0031] Another embodiment involves an disposable scent-diffusing apparatus having a cage including a closed end that defines a plurality of vents, and a perforated end opposite the closed end; a base including an open end and a closed end opposite the open end, wherein the perforated end of the cage and the open end of said base are permanently joined together to create a housing having an interior chamber between the closed end of the cage and the closed end of the base; a scent-diffusing cartridge positioned within the interior chamber of the housing; and a thermoplastic

film positioned within the interior chamber of the housing, between the closed end of the cage and the scent-diffusing cartridge. The scent-diffusing cartridge is adapted for non-active, dry air scent diffusion in which scent molecules can be passively emitted in non-droplet form. The scent-diffusing cartridge is selectively isolatable from and exposed to an exterior environment surrounding the apparatus, wherein the scent-diffusing cartridge is isolated from the exterior environment when the apparatus is in a first position, and wherein the scent-diffusing cartridge is exposed to the exterior environment when the apparatus is in a second position.

[0032] Other embodiments of the scent dispenser/absorber apparatus include a QR code (e.g., on the base of the apparatus). The QR code Q is configured for the purpose of providing access to instructions, advertising, and/or other information.

[0033] Still another embodiment involves an apparatus for diffusing scent into an external environment, including a housing having a first end adapted for communication with the external environment, a second end being out of communication with the external environment, and an interior chamber between the first and second ends and surrounded by an exterior surface of the housing; a scent-diffusing cartridge positioned within the interior chamber of said housing; and a QR code placed on the exterior surface of the housing at a location where it is visible from the external environment.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0034] For a better understanding of the present invention, reference is made to the following detailed description of various exemplary embodiments considered in conjunction with the accompanying drawings, in which:

[0035] FIG. 1 is a perspective view of a scent dispenser/absorber in an “off” condition, the scent dispenser/absorber being constructed in accordance with an exemplary embodiment of the present invention;

[0036] FIG. 2 is an exploded perspective view of the scent dispenser/absorber of FIG. 1;

[0037] FIG. 3 is a perspective view of the scent dispenser/absorber of FIGS. 1 and 2 in an “on” condition;

[0038] FIG. 4 is an exploded perspective view of the scent dispenser/absorber of FIG. 3;

[0039] FIG. 5 is a side elevational view of another embodiment of a scent dispenser/absorber constructed in accordance with the present invention, the scent dispenser/absorber being shown in FIG. 5 in an “off” condition;

[0040] FIG. 6 is a cross-sectional view of the scent dispenser/absorber of FIG. 5, the cross-section being taken through section line 6-6 in FIG. 5 and looking the direction of the arrows;

[0041] FIG. 7 is an exploded perspective view of the scent dispenser/absorber of FIGS. 5 and 6;

[0042] FIG. 8 is a side elevational view of the scent dispenser/absorber of FIGS. 5-7, the scent dispenser/absorber being shown in FIG. 8 in an “on” condition;

[0043] FIG. 9 is a cross-sectional view of the scent dispenser/absorber of FIG. 8, the cross-section being taken through section line 9-9 in FIG. 8 and looking the direction of the arrows;

[0044] FIG. 10 is an exploded perspective view of the scent dispenser/absorber of FIGS. 8 and 9;

[0045] FIG. 11 is an exploded view, shown in perspective, which schematically illustrates various steps in the manufacture of a scent-dispensing insert (i.e., subcombination) and an end product incorporating same (i.e., combination), both of which are representative of other aspects of the present invention;

[0046] FIG. 12 is a partial exploded view, shown in perspective, of the end product produced by the manufacturing steps illustrated schematically in FIG. 11;

[0047] FIG. 13 is a perspective view of a package useful for storing and shipping scent cartridges,

such as those used in connection with the various embodiments of the present invention illustrated in FIGS. **1-12**;

[0048] FIG. **14** shows a further embodiment of a scent dispenser/absorber constructed in accordance with yet another aspect of the present invention, the scent dispenser/absorber of FIG. **14** being shown in an “off” condition;

[0049] FIG. **15** shows the scent dispenser/absorber of FIG. **14** in a fully “on” condition;

[0050] FIG. **16** shows the scent dispenser/absorber of FIGS. **14** and **15** in a partially “on” condition;

[0051] FIG. **17** shows a kit, including a plurality of scent dispensers/absorbers arranged on a stand with some of the scent dispensers/absorbers being shown in “on” conditions and some of the scent dispensers/absorbers being shown in “off” conditions;

[0052] FIG. **18** is a top perspective view of an alternate scent cartridge made in accordance with an embodiment of the present invention;

[0053] FIG. **19** is a top plan view of the scent cartridge of FIG. **18**;

[0054] FIG. **20** is a lateral cross-sectional view of the scent cartridge of FIG. **19** taken along section line **20-20**;

[0055] FIG. **21** is a top perspective view of another alternative scent cartridge made in accordance with an embodiment of the present invention;

[0056] FIG. **22** is a top plan view of the scent cartridge of FIG. **21**;

[0057] FIG. **23** is a lateral cross-sectional view of the scent cartridge of FIG. **22** taken along section line **23-23**;

[0058] FIG. **24** is a graph illustrating diffusion curves in accordance with an embodiment of the present invention utilizing a particular scent;

[0059] FIG. **25** is a graph illustrating diffusion curves in accordance with an embodiment of the present invention utilizing a different scent;

[0060] FIG. **26** is a graphical summary characterizing the performance of various hydrophilic scented materials relative to their pore sizes and densities;

[0061] FIG. **27** is a graph illustrating release of scent over time for different membrane materials in accordance with embodiments of the present invention;

[0062] FIG. **28** is a perspective view of a scent dispenser/holder combination, the scent dispenser being shown in an “on” position in accordance with an embodiment of the present invention;

[0063] FIG. **29** is a side perspective view of the scent dispenser/holder combination shown in FIG. **28**;

[0064] FIG. **30** is a perspective view of the scent dispenser/holder combination of FIGS. **28** and **29**, the scent dispenser being shown in an “off” position in accordance with an embodiment of the present invention;

[0065] FIG. **31** is a perspective view of an alternate scent dispenser/holder combination in accordance with an embodiment of the present invention;

[0066] FIG. **32** is an exploded view of a scent dispenser/absorber constructed in accordance with an exemplary embodiment of the present invention, the scent dispenser/absorber being shown in an “on” (i.e., scent dispensing/absorbing) condition or position;

[0067] FIG. **33** is a front perspective view of the scent dispenser/absorber of FIG. **32** in its “on” condition;

[0068] FIG. **34** is a plan view, looking from the front, of the scent dispenser/absorber of FIG. **33**;

[0069] FIG. **35** is a rear perspective view of the scent dispenser/absorber of FIG. **32** in an “off” (i.e., non-dispensing/non-absorbing) condition or position;

[0070] FIG. **36** is a side elevational view of the scent dispenser/absorber of FIG. **35**;

[0071] FIG. **37** is a plan view, looking from the front or rear, of the scent dispenser/absorber shown in FIGS. **32-36**;

[0072] FIG. **38** is an exploded view of a scent dispenser/absorber constructed in accordance with

another exemplary embodiment of the present invention;

[0073] FIG. **39** is a cross-sectional side view of the scent dispenser/absorber of FIG. **38**, the scent dispenser/absorber being shown on a support surface in an “off” (i.e., non-dispensing/non-absorbing) condition or position;

[0074] FIG. **40** is a plan view, looking from above, of the scent dispenser/absorber of FIG. **39**;

[0075] FIG. **41** is a cross-sectional side view of the scent dispenser/absorber of FIG. **38**, the scent dispenser/absorber being shown on a support surface in an “on” (i.e., scent dispensing/absorbing) condition or position; and

[0076] FIG. **42** is a plan view, looking from above, of the scent dispenser/absorber of FIG. **41**.

DETAILED DESCRIPTION OF VARIOUS EXEMPLARY EMBODIMENTS

[0077] Reference will now be made to the various embodiments of the present invention illustrated in FIGS. **1-31**. Whenever specified hereinbelow, similar or like reference numbers (increased by, for example, one hundred, two hundred, etc.) may be used in the figures to indicate similar or like structure and/or functionality. It should be understood that the figures depict the various embodiments of the present invention for purposes of illustration only, with the further understanding that some illustrated features are depicted schematically due, for instance, to scaling constraints. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated in FIGS. **1-31** may be employed, without departing from the objects of the invention described hereinabove.

[0078] All terms defined herein should be afforded their broadest possible interpretation, including any implied meanings as dictated by a reading of this specification, as well as any words that a person having skill in the art and/or a dictionary, treatise, or similar authority would assign thereto. For instance, the terms, “for example”, “e.g.”, “optionally”, as used herein, are intended to be used to introduce non-limiting examples, while the phrases “in one embodiment” and “in some embodiments” as used herein do not necessarily refer to the same embodiment(s), although they may. Furthermore, the phrases “in another embodiment” and “in some other embodiments” as used herein do not necessarily refer to a different embodiment, although they may. Thus, as described below, various embodiments of the invention may be readily combined, without departing from the scope or spirit of the invention.

[0079] In addition, as used herein, the term “or” is an inclusive “or” operator, and is equivalent to the term “and/or,” unless the context clearly dictates otherwise. The term “based on” is not exclusive and allows for being based on additional factors not described, unless the context clearly dictates otherwise. Throughout the specification, the meaning of “a,” “an,” and “the” includes plural references. The terms “comprises” and “comprising” when used herein specify that certain features are present in that embodiment; however, this phrase should not be interpreted to preclude the presence or inclusion of additional steps, operations, features, components, and/or groups thereof.

[0080] With the foregoing prefatory comments in mind, reference is now made to FIG. **1**, showing a scent dispenser/absorber **10** in a closed or “off” condition, in which a lid **12** is securely fixed to container (i.e., housing) **14**. As shown in FIG. **1**, the lid **12** can be secured and removably attached to an open end **16** of container **14** in a first orientation (i.e., a closed position), which coincides with the “off” condition of the scent dispenser/absorber **10**. In the exemplary embodiment illustrated in FIGS. **1** and **2**, as well as in FIGS. **3** and **4**, container **14** has external threads **11** on its open end **16** (see FIGS. **2** and **4**), while lid **12** has complementary mating internal threads **13** on a skirt **15** of the lid **12** (see FIG. **4**) to securely join the lid **12** to the container **14** via a threaded connection (i.e., the lid **12** and container **14** are threadedly attached to one another by threads **11**, **13**). However, the joinder of the lid **12** and the container **14** may be accomplished via any number of known alternative joining methods, such as dimples, snap fits, press fits, notches, friction fits, pins, magnets, etc. Lid **12** and/or container **14** can be made from glass, polypropylene or related materials with similar properties. As will be described hereinafter, the lid **12** and container **14** can

be separated and then joined in a second (i.e., inverted) orientation. To increase enjoyment, the container **14** and/or lid **12** can have a variety of amusing colors, shapes and designs (see, for instance, FIG. **17**).

[0081] Referring now to FIG. **2**, an exploded view of the scent dispenser/absorber **10** is shown. Specifically, the parts are ordered to show their orientation when scent dispenser/absorber **10** is in its “off” condition. A cage **18** is configured to receive scent cartridge (i.e., insert) **20** inside an internal or interior space (i.e., compartment) **22**, whereby cage **18** secures and protects the scent cartridge **20**. In an embodiment, the cage **18** includes an open end defined by a surrounding, apertured flange **24**, the function of which will be described hereinbelow.

[0082] When the scent dispenser/absorber **10** is in its “off” condition (i.e., with lid **12** threadedly attached to the open end **16** of container **14**), both the cage **18** and the scent cartridge **20** are confined, in an airtight manner, within an inner chamber **26** of the container **14**. The inner chamber **26** is sized and shaped such that the scent dispenser/absorber **10** is sufficiently spacious for receiving the cage **18** and the scent cartridge **20** of specified dimensions. When the scent dispenser/absorber **10** is in its “off” condition, the scent cartridge **20** is cut-off (i.e., isolated) from the external environment by cap (i.e., cover) **27**, whereby scents emanating from scent cartridge **20** cannot escape from the inner chamber **26** of container **14**. The cap **27** also functions as a mounting surface for the flange **24** of the cage **18**, which is therefore carried by the lid **12**. When the scent dispenser/absorber **10** is configured to operate as an absorber of scents, undesirable odors will be unable to be absorbed by scent cartridge **20** as long as the lid **12** is in its closed position and the scent dispenser/absorber **10** is therefore in its “off” condition.

[0083] Referring now to FIGS. **3** and **4**, the scent dispenser/absorber **10** is shown in its “on” condition. Switching the scent dispenser/absorber **10** from its “off” condition to its “on” condition can be accomplished by removing the lid **12** and inverting it, whereby the cage **18** and the scent cartridge **20** project outwardly from the lid **12** and externally of the container **14**. To position the lid **12** on the open end **16** of the container **14**, the lid **12** is provided with a skirt-like ring **28**, which functions to center the lid **12** on the open end **16** of the container **14** where it movably sits when the scent dispenser/absorber **10** is in its “on” condition.

[0084] When the scent dispenser/absorber **10** is in use in its “on” condition, the scent cartridge **20** communicates with the outside environment via a plurality of vents **30** in the cage **18**. In a scent-dispensing mode, the vents **30** allow scent to migrate from the scent cartridge **20** to the surrounding environment. Conversely, in a scent-absorbing mode, the vents **30** would allow scent to migrate from the surrounding environment to the scent cartridge **20**.

[0085] When not in use, the scent dispenser/absorber **10** can be manually switched from its “on” condition back to its “off” condition by removing the lid **12**, inverting it and then reattaching the lid **12** to the container **14** via the aforementioned threaded connection or any other suitable form of connection known in the art. In such an “off” condition, the cage **18** and the scent cartridge **20** are positioned entirely within the inner chamber **26** of the container **14** between the open end **16** thereof and a closed end **32** thereof.

[0086] In some embodiments, the lid **12** removably receives the cage **18** by, for instance, a bayonet-type connection, whereby the cage **18** can be removed to expose the open end **24** of cage **18** and to therefore permit removal or replacement of the scent cartridge **20** and/or the cage **18**. In an embodiment, the scent cartridge **20** is removable and insertable through the open end of the cage **18** (i.e., the end defined by the apertured flange **24**). In an alternate embodiment, the cage **18** is permanently attached to lid **14** by, for instance, a suitable adhesive or any other mechanism known in the art. In further embodiments, the cage **18** can have a door (not shown), whereby the cage **18** can be manually opened for removal or replacement of the scent cartridge **20**. In a further embodiment, the cage **18** is constructed from separable halves (not shown) such that the cage **18** can be opened by separating its two halves (e.g., via a plurality of joints or the like). In the various embodiments and configurations described herein, the cage **18** can be made from glass,

polypropylene or related materials with similar properties.

[0087] Another exemplary embodiment of the present invention is illustrated in FIGS. 5-10. As an initial observation, it is noteworthy that the embodiment of FIGS. 5-10 incorporates a number of structural features similar, if not identical, to counterparts in the embodiment of FIGS. 1-4. In view of such similarities, and in order to facilitate consideration and discussion of the embodiment shown in FIGS. 5-10, structural features of such embodiment that are similar or identical to their counterparts in the embodiment of FIGS. 1-4 will be identified using the same reference number increased by one hundred. It should also be noted that any such counterpart feature of the embodiment shown in FIGS. 5-10 will perform the same function as its corresponding feature in the embodiment of FIGS. 1-4, unless otherwise stated.

[0088] Referring now to FIGS. 5-7, a scent dispenser/absorber **110** is shown in a closed or “off” condition, in which lid **112** is secured and removably attached to an open end **116** of container (i.e., housing) **114** in a first orientation (i.e., a closed position), which coincides with the “off” condition of the scent dispenser/absorber **110**. In the exemplary embodiment illustrated in FIGS. 5-7, container **114** has external threads **111** on its open end **116** (see FIGS. 6 and 7), while lid **112** has complementary mating internal threads **113** on skirt **115** of the lid **112** (see, for instance, FIG. 6) to securely join the lid **112** to the container **114** via the threaded connection created by the threads **111**, **113**. However, the same form of joinder may be accomplished via any number of known joining methods, such as dimples, snap fits, press fits, notches, friction fits, pins, magnets, etc. Lid **112** and/or container **114** can be made from glass, polypropylene or related materials with similar properties. As will be described hereinafter, the lid **112** and the container **114** can be separated and then joined in a second (i.e., inverted) orientation. To increase enjoyment, the container **114** and/or lid **112** can have a variety of amusing colors, shapes and designs.

[0089] With particular reference to FIGS. 6 and 7, in which the scent dispenser/absorber **110** is shown in its “off” condition, lid **112** includes a cap (i.e., cover) **127** from which the skirt **115** extends in a first axial direction relative to imaginary axis “A” (see FIG. 7). Another skirt **117** extends from the cap **127** of the lid **112**, but in a second axial direction opposite the first axial direction. In other words, the skirts **115**, **117** extend from the cap **127** of the lid **112** in opposite axial directions.

[0090] As can be seen in FIG. 6, when the scent dispenser/absorber **110** is in its “off” condition, the external threads **111** on the open end **116** of container **114** are threadedly engaged with the internal threads **113** on the skirt **115** of the lid **112**. Similar internal threads **119** are provided on the skirt **117** of the lid **112** for a purpose to be described hereinafter. The cap **127** of the lid **112** is also provided with a mounting rim **121** having an annular lip **123** projecting radially outwardly from the mounting rim **121** for a purpose to be described hereinafter.

[0091] A domed and apertured diffuser **125** is releasably attached to the mounting rim **121** of the lid **112**. More particularly, and with particular reference now to FIG. 6, the diffuser **125** has an annular trough **129** sized and shaped such that the trough **129** can be releasably engaged by the lip **123** on the mounting rim **121** of the lid **112**. Apertures **131** are provided in the diffuser **125** for a purpose to be described hereinafter. As shown in FIG. 6, the diffuser **125** and, hence, the apertures **131** therein project downwardly into inner chamber **126**, when the scent dispenser/absorber is in its “off” condition.

[0092] With continued reference to FIGS. 6 and 7, the lid **112** and diffuser **125** cooperate to form a compartment **133** (see FIG. 6) sized and shaped to receive a scent cartridge **120**. When the scent dispenser/absorber **110** is in its “off” condition as shown in FIGS. 5-7, the scent cartridge **120** is confined, in an airtight manner, within an inner or interior chamber **126** of the container **114**. The airtight connection between the lid **112** and container **114** is facilitated by the provision of an O-ring (i.e., gasket) **135**, which is sandwiched between the open end **116** of the container **114** and the cap **127** of the lid **112**. When the scent dispenser/absorber **110** is in its “off” condition, the scent cartridge **120** is cut-off from the external environment by the cap **127** of the lid **112**, whereby scents

emanating from scent cartridge **120** cannot escape from the inner chamber **126**.

[0093] With continued reference to FIG. **6**, the inner chamber **126** is adapted (i.e., sized and shaped) to house a flameless, scent-free imitation candle **137**, the aesthetics of which would be available with or without the functional benefits of dispensing scent. Such aesthetics can be achieved by mounting a light source (i.e., a light-emitting diode “LED”) **139** on closed end **132** of the container **114**. For increased immersion, the light source **139** can be projected through a flame-shaped globe **141**, which can be colored such that globe **141** imitates a real flame. In one embodiment, the container **114** can be partially or fully transparent to allow light emitted from the light source **139** to radiate outwardly. A remote or manually-operated switch **143** can control the operation (i.e., the “on” and “off” states) of light source **139**, which may function as a night light, for example. In an alternate embodiment, the light source **139** can be located outside container **114**. In other alternate embodiments, decorative modifications, such as a tinted paper filter or decorative patterns (not shown) can be placed as follows: around the light source **139**; inside the container **114**; and/or on the exterior of the scent dispenser in order to emit different colors and/or patterns of light. These optional patterns, etc. could be permanent or adapted to be interchanged with other decorative elements. For instance, a Halloween theme could be adopted by providing a pattern with Jack-o-lanterns, witches, etc. In one embodiment, such patterns and the like can be interchanged by simply unscrewing the lid **112**, removing one pattern and replacing it with another (e.g., a pattern with a Thanksgiving theme could be provided subsequent to the use of a pattern having a Halloween theme).

[0094] It should be noted that the inner chamber **126** of the container **114** is configured such that the scent cartridge **120** and/or a number of replacement cartridges (not shown) can be stored within the container **114** during shipping or storage, for instance. It should also be noted that when the scent dispenser/absorber **110** is configured to operate as an absorber of scents, undesirable odors will be unable to be absorbed by scent cartridge **120** as long as the lid **112** is in its closed position and the scent dispenser/absorber **110** is therefore in its “off” condition.

[0095] Referring now to FIGS. **8-10**, the scent dispenser/absorber **110** is shown in its “on” condition. Switching the scent dispenser/absorber **110** from its “off” condition to its “on” condition can be accomplished by removing the lid **112** and inverting it, whereby the diffuser **125** and the scent cartridge **120** project outwardly from the lid **112** and externally of the container **114**. When the scent dispenser/absorber **110** is in its “on” condition, the lid **112** is securely attached to the open end **116** of the container **114** via the engagement of threads **111**, **119**, while the scent cartridge **120** communicates with the outside environment via the apertures **131** in diffuser **125**. In a scent-dispensing mode, the apertures **131** allow scent to migrate from the scent cartridge **120** to the surrounding environment. Conversely, in a scent-absorbing mode, the apertures **131** would allow scent to migrate from the surrounding environment to the scent cartridge **120**.

[0096] When not in use, the scent dispenser/absorber **110** can be manually switched from its “on” condition back to its “off” condition by removing the lid **112**, inverting it and then reattaching the lid **112** to the container **114** via the engagement of threads **111**, **113**. In such an “off” condition, the diffuser **125** and the scent cartridge **120** are positioned entirely within the inner chamber **126** of the container **114** between the open end **116** thereof and the closed end **132** thereof.

[0097] The scent dispenser/absorber **110** is especially adapted for use in combination with a cupholder in a land, sea or air vehicle. So as to adapt the scent dispenser/absorber **110** for such uses, the container **114** can be sized and shaped to be received within common cupholders, the size and shape of the container **114** comporting with the associated dimensions of such cupholders. In connection with such uses, the lid **112** of the scent dispenser/absorber **114** will be securely attached to the container **114** due to the threaded connection created by threads **111**, **113**, **119**, whether the scent dispenser/absorber **110** is in its “on” condition or its “off” condition.

[0098] As also illustrated in FIG. **10**, certain embodiments of the scent dispenser/absorber **110** include a QR code Q on the domed surface of the diffuser **125**. The QR code Q is configured for

the purpose of providing access to instructions, advertising, and/or other information. While the QR code is shown as being placed on the domed surface of the scent dispenser/absorber **110**, the QR code may be placed elsewhere on the scent dispenser/absorber **110** in other embodiments.

[0099] The scent cartridges **20**, **120** may be made of foam or any other suitable material (e.g., reticulated hydrophilic polyurethane foam composite, preferably about $\frac{3}{8}$ of an inch in thickness) adapted to absorb or adsorb a volatile scented substance. In the exemplary embodiments that follow, the scent cartridges **20**, **120** perform scent dispensing functions using evaporation to emit scent molecules as “dry air” (i.e., without droplets). In such “dry air” embodiments, the scent cartridges **20**, **120** are infused with odorant such that odorant molecules spread throughout the surface of the material of the scent cartridges **20**, **120**. With the scent dispenser/absorbers **10**, **110** in their “off” conditions (see FIGS. **1** and **2** and FIGS. **5-7**, respectively) evaporation of the odorant molecules will be contained within the containers **14**, **114**, respectively, thereby filling the inner chambers **26**, **126** around the scent cartridges **20**, **120**, respectively. In such a condition, the scent dispenser/absorbers **10**, **110** are adapted for storage and/or transportation.

[0100] In one embodiment, the scent cartridges **20**, **120** comprise reticulated hydrophobic polyurethane foam (Product No. RT030CHRSC1) from Ionac (Woodbridge INOAC Technical Products, 100 Carol Place, Moonachie, NJ 07074), which material provides the scaffolding for the scent cartridges **20**, **120**. It is a desirable material for the scaffolding of the scent cartridges **20**, **120** because scent molecules only weakly interact with it. Due to the fact that polyurethane molecules are not volatile at typical pressures and temperatures, the odorant scent released from the scent cartridges **20**, **120** into the inner chambers **26**, **126** of the containers **14**, **114**, respectively, will not be adulterated by the scaffolding (i.e., the aforementioned polyurethane foam).

[0101] To prepare the scent cartridges **20**, **120**, the foam or scaffolding material can be coated with a thin layer of prepolymer, such as Dow Chemical's HYPOL JM 5005, which can then be reacted with water molecules to form a hydrophilic polyurethane surface coating. The large hydrophilic polyurethane molecules are non-volatile and do not contribute to the scents produced. The hydrophilic polyurethane is polar in nature and has a degree of affinity for scent molecules.

[0102] As the hydrophilic polyurethane surface coating is curing, CO.sub.2 out gassing results in a larger surface area. On the surface of the reticulated material, the scent adheres along the hydrophilic polyurethane surface due to their polar interactions. There is a large surface area available for evaporation and a “headspace” within the material that generally contains a high level of “dry air” scented vapor. Through this process, the scent cartridges **20**, **120** can release a high quality representation of the original scent, while ensuring that the scent lasts a long time. Various thicknesses (e.g., $\frac{3}{8}$ ”) and pore sizes (e.g., 30 pores per inch) of reticulated base material are options.

[0103] The result is that when the scent dispenser/absorbers **10**, **110** are switched to their “on” conditions, as described above, a faithful iteration of the original odorant scent will be released for sampling or other purposes when used in conjunction with scent cartridges **20**, **120**. In other words, the aforementioned sampling step is carried out in an environment in which essentially ambient air contains scent molecules, but no liquid. Thus, the scent dispenser/absorbers **10**, **110** allow scents to be stored, transported and sampled in a non-liquid form, the subsequent sampling being performed in a “passive” manner.

[0104] Furthermore, when coupled with the ability of the presently proposed scent dispensers (e.g., the scent dispenser/absorbers **10**, **110**) to be sealed when not in use, a longer effective scent life can be achieved. This is in addition to the fact that the inventive scent inserts (e.g., the scent cartridges **20**, **120**) also affect the scent diffusion curve, which would further advance the goal of increasing the life of the scenting function of their associated scent dispensers.

[0105] While the scent cartridges **20**, **120** have been described with reference to a particular embodiment thereof, it should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the

present invention. For instance, the scent dispenser/absorbers **10**, **110** are adapted for use with other types of scent capturing and diffusing media, such as the scent cell disclosed in U.S. Patent Application Publication No. 2017/0312380 (see especially Paragraphs [0048] to) [0050]U.S. Pat. No. 8,544,766 (see especially Column 6, lines 30-59) and U.S. Pat. No. 6,617,014, all three of which patent publications are incorporated herein by reference in their entireties. In addition, the scent cartridges **20**, **120** can be replaced by, for example, a scent cartridge adapted to absorb and/or adsorb a volatile scented substance, while allowing the ready passage of air therethrough. In some embodiments, the scent cartridges **20**, **120** would contain an amount of volatile scented substance, such as those used in perfumes, as well as those used for attracting game, for calming a pet, for functioning as a diet aid, for aroma therapy, for medical applications, or for other uses which are known or may become known, such as providing cannabis-derived and/or hemp-derived scents, fragrances, etc. In other embodiments, the scent cartridges **20**, **120** would be designed such that the scented substance may be added directly to the scent cartridges **20**, **120** to replenish or change the scent.

[0106] By way of further example, the scent cartridges **20**, **120** may be made of any material that can carry and release volatile scented substances. In some embodiments, it could be made of an absorbent fibrous material or closed cell foam having air passages penetrating therethrough. In other embodiments, the scent cartridges **20**, **120** could be made of an open-cell foam that presents an appreciable ratio of surface area to volume of foam, with higher ratios typically being preferred. In such embodiments, the foam may be a hydrophilic foam or have a hydrophilic material exposed at the surfaces of its cells. Other suitable embodiments could utilize an open-cell foam composite made of substantially hydrophobic foam to provide structure to the composite and substantially hydrophilic foam exposed at the surfaces of its cells, such as the foam described in U.S. Pat. No. 6,617,014, whose disclosure in that regard is incorporated herein by reference.

[0107] In yet other embodiments, the scent cartridges **20**, **120** may comprise a nonwoven fibrous material substrate coated with, for example, a substantially hydrophilic foam coating which is exposed at the surface and in interstitial spaces within the nonwoven fibrous material. The interstitial spaces within the nonwoven fibrous material form air passages penetrating therethrough to allow the flow of air. Examples of suitable nonwoven fibrous materials include, without limitation, cotton, felt, silk, or combinations thereof. As will be recognized by persons of ordinary skill in the relevant art, such embodiments would be useful when the volatile scented substances employed to impart scent or alternative odor to the scent cartridges **20**, **120** are of the types that may react with and degrade some hydrophobic foams (see, for example, U.S. Pat. No. 8,544,766, whose disclosure in that regard is incorporated herein by reference).

[0108] One possible process for producing a non-woven, fibrous scent cartridge involves contacting a substrate of nonwoven fibrous material with a prepolymer emulsion and then polymerizing or curing the emulsion. By way of further example, the substrate can be dipped or immersed in the prepolymer emulsion, which can also be applied by brushing or otherwise coating onto the substrate. In an embodiment of such a process, the substrate of nonwoven fibrous material may be provided as a sheet or block and then coated with the prepolymer emulsion, followed by polymerization or curing of the emulsion to form the substantially hydrophilic foam on the nonwoven fibrous substrate. The substrate can then be cut into appropriately sized and shaped pieces to produce a scent cartridge (e.g., scent cartridge **20** or scent cartridge **120**) suitable for use with its associated scent dispenser (e.g., the scent dispenser/absorber **10** or the scent dispenser/absorber **110**).

[0109] Suitable scent cartridges can be constructed in accordance with still further embodiments thereof. For instance, and with reference to FIGS. **11** and **12**, another alternative scent cartridge arrangement is illustrated, along with a method of making same. With particular reference to FIG. **11**, there is shown a series of illustrations corresponding to various steps in the manufacture of a scent-dispensing insert **210** and a product **212** incorporating the insert **210**. More particularly, and

referring still to FIG. **11**, the initial step in the manufacturing process involves providing a reticulated, hydrophilic composite material **214**, such as a foam composite made up of a scaffold of an open cell hydrophobic material having a plurality of surfaces defining a plurality of pores (e.g., in a range of from about thirty pores per inch to about fifty pores per inch), and a coating of a substantially hydrophilic foam material disposed upon the surfaces of the hydrophobic foam. The resulting foam composite exhibits the structural characteristics of the hydrophobic foam and the absorbency characteristics of the hydrophilic foam. Other suitable materials are disclosed in, for example, U.S. Pat. No. 6,617,014, whose entire disclosure is incorporated herein by reference, as well as in U.S. Pat. No. 10,912,855, whose entire disclosure is incorporated herein by reference. The size and shape of the composite material **214** would be dependent upon its intended end use, examples of which will be discussed hereinafter.

[0110] While not depicted in FIG. **11**, the next step in the manufacturing process involves loading, at least partially, an appropriately sized and shaped piece of the aforementioned composite material **214** with a liquid odorant. By way of example, the liquid odorant can be a volatile scented substance, such as those identified hereinabove, as well as those useful for aroma or insomnia therapy, for medical diagnostic applications and/or for animal husbandry (i.e., pet calming) purposes by, for instance, having the scent emulate that of the pet owner.

[0111] Returning now to FIG. **11**, the next step in the manufacturing process involves enveloping, encasing, enwrapping or otherwise enclosing the scented composite material **214** in a heat-sealable, thermoplastic membrane or film **216** having a permeability that allows scent molecules originating from the odorant to be released from the composite material in a passive and continuous (i.e., without manual or mechanical manipulation) manner. In use in combination with the composite material **214**, the thermoplastic membrane or film **216** performs a metering function in that it prevents, or at least inhibits, the release of liquid, while permitting the release of scent molecules, thereby delivering scent to a user in a dry format free from mists, sprays, aerosols and liquid. To facilitate their combination, the scented composite material **214** and the thermoplastic membrane or film **216** can be shaped to complement one another.

[0112] Suitable thermoplastic materials include, but are not limited to, the following copolymers: high density polyethylene (HDPE); low density polyethylene (LDPE); polypropylene (PP); and ethylene vinyl acetate (EVA), all of which would have a thickness determined by their intended end use. In an embodiment, the heat-sealable thermoplastic membrane has a thickness of about 3 millimeters to about 6 millimeters. In a preferred embodiment, adsorbency of at least one unit of weight of scent per one unit of weight of core material is achieved.

[0113] One specific example of a suitable thermoplastic film is 3M's CoTran Ethylene Vinyl Acetate Membrane Film 9712, 18.5% VA, 2 mil, translucent, no corona treatment. Other suppliers of suitable thermoplastic materials for use in practicing the present invention include, but are not limited to, the following companies: Bemis Company; Berry Plastics; Sigma Plastics; and Sealed Air Corporation. Regardless of the supplier and type of thermoplastic film that is utilized, the resulting combination (i.e., the scent-dispensing insert **214**) includes an inner core, which is in the form of the scent-laden composite material **214**, and an outer skin, which is in the form of the thermoplastic membrane or film **216** heat-sealed around the inner core.

[0114] The thermoplastic membrane or film **216** can have an opaque surface (not shown) capable of being printed on for advertising and/or informational purposes, for instance. In other embodiments, adding the heat-sealable, thermoplastic membrane to the core material enables the placement of imagery or other content on the surface of the product associated with the scent contained therein. By exploiting the association of scent with images, an enhanced sensory experience can be provided to the user.

[0115] The resulting combination, in addition to carrying the desired scent in a non-liquid state for use in, for example, the scent dispensers **10**, **110**, achieves multiple advantages. For instance, the combination of the foam composite and the thermoplastic membrane achieves a synergy that

further increases the lifetime of the scented product. Both the foam composite and the membrane reduce evaporation rates. The effective lifetime, as a function of the capacity of the scented material, therefore increases.

[0116] The foam composite holds and releases scent molecules over time, doing so while reducing the effective evaporation rate of the scent and lengthening the duration of release relative to other passive diffusion methods that do not utilize the foam composite material of the present invention. The foam composite also serves to physically retain the scent, helping to prevent leaks and spills. The thermoplastic membrane also contributes to preventing leaks and/or spills.

[0117] The inventive combination also does not allow odorant oils in the substrate to wick out because the thermoplastic membrane functions to cause the scent molecules to permeate through the tortuous paths in the non-porous (i.e., solid) film structure membrane. This functionality of the thermoplastic membrane allows scent molecules to be discharged (i.e., diffused) therefrom in gaseous form as dry, but scented, air. In some embodiments, the thermoplastic membrane would stop any scent-associated liquids from escaping, even when the inventive scent insert is dropped.

[0118] In use, the heat-sealable, thermoplastic membrane creates an additional barrier for the scent molecules. While one side (i.e., the exterior side) is open to the atmosphere and has a low concentration of scent molecules, the other side (i.e., the interior side) has a much higher concentration of scent molecules. Provided the evaporation rate is high relative to the diffusion of scent molecules through the membrane, there will be minimal residual scent adhering to the membrane itself.

[0119] A further advantage of the present invention is that the thermoplastic membrane can be overfilled, either accidentally or intentionally, while still not permitting scent to escape. Additionally, the presently proposed invention reduces the amount of scenting liquid needed. This reduces the amount of liquid the user needs to interact with, while also minimizing the weight and cost of associated product shipments (e.g., in connection with the library concept disclosed herein). In some embodiments, replacement scent can be provided in 1 oz bottles weighing as little as 0.75 grams.

[0120] The combination of a reticulated hydrophilic polyurethane foam composite heat sealed in an EVA thermoplastic membrane has notable advantages. Typically, EVA is used to meter scents in their liquid or solid states by using EVA beads or EVA bags of fluid. However, when used in combination with the foam as presently proposed, the scent is evaporated within the EVA enclosure, thereby creating a condition that provides a concentration of scent on the inside of the EVA that is similar to the vapor pressure of the scent being delivered. This condition (i) controls the rate of diffusion relative to the evaporation rate of the scent, thereby enabling longer durations of use, and (ii) minimizes the amount of scent adhering to the outer (i.e., exterior) surface of the membrane. Among other effects, this condition enables manual handling of the scented product or combination (e.g., the scent cartridges **20**, **120**), while leaving minimal residual scent on a user's hands, clothes or any other surfaces the EVA may be placed in contact with.

[0121] A scent that has a lower vapor pressure would diffuse in its liquid state through an EVA bead or bag until a substantial film builds up on the outer surface to slow or halt diffusion. In contrast, the foam of the present invention would adsorb liquid scent, and the rate of diffusion through the thermoplastic film would be limited only by what could already evaporate from the foam matrix at a given temperature; consequently, the rate at which molecules reach the surface of the thermoplastic membrane would be similar to the rate at which they are evaporating from the surface. Additionally, there is no surface film on, for instance, the scent dispensing insert **210**.

[0122] For the particular end use disclosed herein (see FIG. **12**), the aforementioned core/skin combination (i.e., the scent-dispensing insert **210**) would be appropriately sized, shaped and scented to function as a scent-dispensing insert for a pillow, cushion or pouch **218** having a fabric covering or casing. Thus, the next step in the manufacturing process would involve placing the scent-dispensing insert **210** inside the fabric-covered pillow, cushion or pouch **218**, where the insert

210 would provide a long-lasting, high-fidelity scent-dispensing experience (see FIGS. **11** and **12**). [0123] As explained above, adding the heat-sealable, thermoplastic membrane to the core material provides an additional containment means for the liquid scent when adsorbed in the foam composite. This effect protects against the escape of scent liquid when the product is, for instance, dropped. This result is desirable for multiple reasons. For instance, customers or other end users can handle scented products without getting an appreciable amount of scent on their skin, clothes, etc. Furthermore, the scented product can be placed on a finished surface (e.g., a table) with minimal to low risk of damaging the finish with scent. This being the case, the present invention can provide diffusion, even while touching finished surfaces, without harming those surfaces. Similarly, the present invention can be used in conjunction with fabrics without wicking out or damaging the fabrics.

[0124] When used in combination with a pillow or cushion (see, for instance, FIG. **12**), the thermoplastic outer skin **216** of the insert **210** can be provided with anti-microbial and/or anti-fungal properties, for hygienic purposes. The insert **210** can simply be placed inside the pillow or cushion **218**; or it can be attached to the pillow's or cushion's fabric covering or casing by sewing, stitching or any other conventional attachment means. In use, scent molecules originating from the odorant are passively and continuously released from the reticulated, hydrophilic inner core **214** of the insert **210**. The scent molecules released from the inner core **214** of the insert **210** pass through the thermoplastic outer skin **216** of the insert **210** and then through the fabric casing of the pillow or cushion **218**, before being diffused into the surrounding environment to provide a faithful iteration of the original odorant or scent. As mentioned above, wicking of scent from the inner core **214** to the fabric of the pillow or cushion **218** is inhibited by the thermoplastic outer skin **216**, which functions as a barrier between the fabric and the inner core. Scent residue on the fabric is to be avoided because it could result in increased diffusion rates and unpleasant feel, while potentially damaging foreign objects coming into contact with such residue. For cleaning and/or sanitizing purposes, the fabric could be of a washable variety.

[0125] When used in combination with a pouch, the size of the insert **210** would be reduced to account for the reduced size of the pouch, which would typically be smaller than that of a pillow or cushion. The pouch could be adapted for releasable attachment to a dog pen, for example, where it would emit a scent (e.g., a scent reminiscent of the pet's owner) selected to have a calming effect on the pet housed in the pen.

[0126] In some embodiments, evaporation rates for personal scenting applications in accordance with the present invention range from about 5 mg to about 25 mg per day. In an embodiment, a scent dispenser made in accordance with an embodiment of the present invention could be specially adapted for use with a desk. Such a dispenser would be desirable to increase productivity, soothe a user and operate as a light source if, for instance, combined with an LED light.

[0127] Similar objectives can be achieved by adapting the inventive device for use in a car or other land, sea or air vehicle. This could entail both driving-related scents, as well as such features as adapting the dimensions of the device to fit into a conventional cupholder. By way of further example, the device, especially when in the form of a pouch, could be adapted for use as a hanging household and/or vehicle air freshener.

[0128] It should be understood that the volume of the foam-like inner core **214** of the insert **210** can be varied to control the effective life and/or the scent diffusion capacity of the insert **210**. The permeability and/or thickness of the thermoplastic outer skin **216** can also be selected to control the effective (i.e., operational) life and/or the scent diffusion capacity of the insert **210**. By way of example, the active life and rate of diffusion may be varied by adjusting membrane/film properties, such as thickness, vinyl acetate concentrations, number of layers, etc. Adding the heat-sealable, thermoplastic membrane to the core material allows for the further control of the rate of evaporation by, for example, adjusting the material or the physical construction of the membrane/film itself. Other factors capable of controlling diffusion rates and capacities include,

but are not limited to, the volume and surface area of foam in conjunction with variations of the surface area of membrane material. These factors enable the device to be better adapted to meet the diffusion needs of the wide variety of scent uses. For EVA specifically, the permeability of typical odorant molecules is generally higher when used with higher concentrations of VA. Certain plastics have negligible permeabilities by comparison. The effects of varying further parameters will be demonstrated in the Examples section below, as well as by the following representative parameters. The thermoplastic membrane may comprise one or more polymers selected from the group consisting of HDPE; LDPE; PP and EVA. When the polymer is EVA, the EVA can be selected such that the EVA has (i) a VA concentration in a range of from about 3.5% to about 18.5% of the EVA, and (ii) a thickness in a range of from about 2 millimeters to about 6 millimeters. When the polymer is HDPE, the HDPE can have a thickness in a range of from about 1 millimeter to about 6 millimeters. When the polymer is LDPE, the LDPE can have a thickness in a range of from about 1 millimeter to about 6 millimeters. The hydrophobic core of the scaffold can have (i) a volume in a range of from about 0.1 cubic inches to about 4 cubic inches (or, more preferably, from about 0.5 cubic inches to about 2.25 cubic inches), and (ii) a weight in a range of from about 150 milligrams to about 6 grams. The hydrophobic core of the scaffold absorbs liquid odorant and releases scent molecules emanating therefrom, the scent molecules being discharged from the scaffold through the hydrophilic coating thereof. The scent molecules emanating from the liquid odorant (e.g., an amount of liquid odorant in a range of from about 50 milligrams to about 12 grams) create a relatively low level of residual scent on the outer surface of the thermoplastic membrane, which level is dependent upon the liquid odorant's evaporation rate (e.g., in a range of from about 5 mg to about 25 mg per day).

[0129] While in some embodiments, such as those disclosed hereinabove, the insert **210** functions as an interior attachment to a useful article, in other embodiments the insert **210** can be adapted to function as an exterior attachment to a useful article, such as various different types of articles worn by a user (e.g., backpacks, jewelry, belts, etc.). The insert **210** can also be used to produce scents designed to improve an individual's memory, as well as a learning experience that an individual may be undertaking (even when such an experience is undertaken during sleep).

[0130] FIGS. **14-16** show a non-invertible scent dispenser/absorber **310**. For example, the scent dispenser/absorber **310** can be transitioned between a fully closed position (i.e., “off” condition) as shown in FIG. **14**, and a fully open position (i.e., “on” condition) as shown in FIG. **15**. In operation, a housing **312** works in telescopic fashion with an interior casing **314**, such that the housing **312** operates as a piston and the casing **314** as a cylinder. To transition between the closed position illustrated in FIG. **14** and the open position illustrated in FIG. **15**, a user can grip the casing **314** and pull it outward from the housing **312**. As the casing **314** is extended in such a telescopic fashion, ports **316** on the casing **314** are completely exposed to the open air, thereby placing scent cartridge **318** in communication with the surrounding environment. While the scent cartridge **318** can be provided as a single integrated part, it may also have a multi-part construction as shown in, for example, FIGS. **15** and **16**, which show a pair of scent cartridges **318a** and **318b** arranged back-to-back within the casing **314**. If the scent cartridges **318a** and **318b** are provided with different scents, any of the ports **316** aligned solely with scent cartridge **318a** would emit one scent, while any of the ports **316** aligned solely with the scent cartridge **318b** would emit a different scent and any of the ports **316** aligned with both of the scent cartridges **318a** and **318b** would emit a customizable, composite (i.e., hybrid) scent. This customizable, scent-mixing can also be employed by other non-invertible scent dispensers, such as those disclosed in the aforementioned U.S. Pat. No. 8,544,766. It should also be noted that the amount of scent diffusion can be limited by positioning the casing **314** such that the ports **316** are partially exposed to the air and partially blocked by the housing **312** (see, for instance, FIG. **16**).

[0131] FIG. **17** shows a pack or kit which includes a rack-like stand **410** configured to store a plurality of scent dispensers/absorbers **412**, some of which are shown in “on” conditions and others

of which are shown in “off” conditions. In use, the various scent dispensers/absorbers **412** can incorporate scents associated with various times of day. Some or all of the scent dispensers/absorbers **412** might even be equipped with means to play sounds associated with the scents to be dispensed.

[0132] In an embodiment, the stand **410** of FIG. **17** and/or similar displays could be used as part of a broader system and/or method to provide desirable scents to a user in the form of a repository (i.e., library) of replaceable and interchangeable scent cartridges (not shown), such as scent cartridges **20**, **120**. In an embodiment, such replaceable and interchangeable scent cartridges could be provided on a per unit basis, or as part of a subscription service, as replacements for existing scent cartridges, or they could contain different scents. In an embodiment, the repository could have a replenishable stock (e.g., a seasonally-varying selection) of interchangeable and/or replaceable scent cartridges, with new scents being provided over time by the repository (i.e., scent library) itself, or by third parties selling their own scent inserts or cartridges through the scent library or otherwise. In an embodiment, the repository could be organized (physically or categorically) into a plurality of sub-categories, each being associated with different activities (e.g., study aide scents, holiday scents, anxiety relief scents, soothing scents, video call scents, meeting scents, desk work scents, etc.). The interchangeable scent cartridges could also tie into the so-called wellness movement. In an embodiment, a subscription service could also be associated with a given sub-category (e.g., providing different scents from the sub-category over time). In another embodiment, interchangeable and replaceable scent cartridges could be supplied to a user randomly (i.e., without the user's prior knowledge of the scents provided). Alternatively, a known scent could be automatically provided to a user on a periodic basis as a “refill.”

[0133] In an embodiment, the replaceable and interchangeable scent cartridges could be provided in combination with replacement cages that would be essentially identical to cage **18** (see, for example, FIGS. **2-4**). The replacement cages of such a combination could be distributed from the scent library together with an associated interchangeable and replaceable scent cartridge (e.g., with the interchangeable and replaceable scent cartridge lodged inside the replacement cage), thereby forming a sub-assembly or subcombination. For a user's convenience, the resulting combination or subcombination could be kept in one piece to be selectively attached to, for instance, the lid **12** of scent/dispenser absorber **10** (see, for example, FIGS. **2-4**).

[0134] In other embodiments, the interchangeable and replaceable scent cartridges would not be associated with a replacement cage **18** and would instead be adapted to be swapped into the cage **18** of a user's previously purchased scent dispenser/absorber **10**. In one embodiment, this could be achieved by removing the cage **18** from the lid **12**, removing the previous scent cartridge **20** or replaceable and interchangeable scent cartridge (if any), optionally disinfecting the cage **18**, inserting a new interchangeable and replaceable scent cartridge **20** and reattaching the cage **18** to the lid **12**. In another embodiment, the interchangeable and replaceable scent cartridge **20** could slide into and out of the cage **18**, or it could be inserted via a door in the cage **18** as described hereinabove.

[0135] The interchangeable and replaceable scent cartridges, whether provided alone or in combination with a replacement cage **18**, could be adapted for use with scent dispenser/absorber **10**, which itself could be used individually or as a member of a family of scent dispensers/absorbers. In an alternate embodiment, a display similar to the rack **410** of FIG. **17** could be used to hold interchangeable and replaceable scent cartridges when they are not in use, provided the individual compartments of the rack are modified to seal scent molecules therein. The rack and/or its compartments could be labeled or have indicia associated with specific scents, so that a user could quickly see where the various scent-specific scent cartridges or cartridge/cage combinations are stored.

[0136] In some embodiments, scent cartridges **20**, **120**, or the interchangeable and replaceable scent cartridges could be used with associated containers. These containers could, for example, be

specially adapted to seal scent molecules inside, thereby maintaining the integrity of the interchangeable and replaceable scent cartridges. In an embodiment, these containers would house the interchangeable and replaceable scent cartridge(s) during transit to an end user or other distribution point, the containers being specially adapted to retain scent molecules while being inert thereto. Furthermore, such shipping containers could serve a dual function by providing the end user with a handy storage mechanism for their interchangeable and replaceable scent cartridges when they are not in use. In use, multiple shipping containers could be adorned stylistically and then placed in a rack/base, leading to an arrangement similar to that shown in FIG. 17. In this manner, a user can accumulate his or her own collection of scents from which they can select at their convenience for use in connection with scent dispenser/absorber 10. Alternatively, an array of scent dispensers/absorbers similar to scent dispenser/absorber 110 could constitute the arrangement of FIG. 17.

[0137] An alternative packaging specially adapted for shipping scent cartridges is shown in FIG. 13, which depicts a resealable foil bag 510 whose inner surface is adapted to be inert to any reaction with respect to scent molecules of, for instance, scent cartridges 20, 120. The foil bag 510 (e.g., the Static Shielding Bag 1000 series sold by SCS) thereby functions to eliminate losses and to preserve the potency of scent for the eventual user. The size and shape of the foil bag 510 can be chosen to accommodate one or more scent cartridges having a variety of different sizes and shapes, such as those depicted in FIGS. 18-20 and 21-23, respectively.

[0138] With reference to FIGS. 18-20, there is shown a scent cartridge or insert 610 having a round, disc-like shape. The scent cartridge 610, like the various scent cartridges disclosed hereinabove, but especially the one disclosed in conjunction with FIGS. 11 and 12, includes an inner core of a reticulated, hydrophilic composite material 612 (see FIG. 20) and an outer skin made from a heat-sealable, thermoplastic membrane 614. In the embodiment depicted in FIGS. 18-20, the thermoplastic membrane 614 is formed by heat-sealing a pair of circle-shaped sheets 616, 618 together such that they form seal 620, while enveloping the composite material 612 in the manner described above in conjunction with FIGS. 11 and 12, for example.

[0139] Referring now to FIGS. 21-23, there is shown a scent cartridge or insert 710 having a rectangular, block-like shape. Like the scent cartridge 610, the scent cartridge 710 includes an inner core of a reticulated, hydrophilic composite material 712 (see FIG. 23) and an outer skin made from a heat-sealable, thermoplastic membrane 714. In the embodiment depicted in FIGS. 21-23, the thermoplastic membrane 714 is formed by heat-sealing a pair of rectangle-shaped sheets 716, 718 together such that they form seal 720, while enveloping the composite material 712 in the manner described above in conjunction with FIGS. 11 and 12, for example.

[0140] FIGS. 28-30 show various views of a scent dispenser/holder combination 810, which includes a scent dispenser 812, which can be loaded with one of the inventive scent cartridges described above, mounted on a stand 814. Stand 814 includes a base 816 and a cradle 818 angled with respect to base 816 (see, especially, FIG. 29). Cradle 818 includes fingers 820 adapted to grip scent dispenser 812 and secure it in place. Scent dispenser 812 has apertures 822 through which scent can be dispensed. In certain embodiments, scent dispenser 812 includes tabs 824 which can, for instance, be used to open the scent dispenser 812 to access its interior, such as for the purpose of changing out the scent cartridge (not shown) housed therein. FIG. 28 shows the scent dispenser 812 in an “on” position, wherein its apertures 822 are exposed to open air and scent can be freely dispensed. On the other hand, FIG. 30 shows the scent dispenser 812 in an “off” position, wherein the scent dispenser 812 is placed face-down on stand 814 by a user, thereby blocking the apertures 822 and preventing the egress of scent.

[0141] As also illustrated in FIG. 28, certain embodiments of the scent dispenser 812 include a QR code Q, which is configured for the purpose of providing access to instructions, advertising, and/or other information. While the QR code is shown as being placed on a bottom surface of the scent dispenser 812, the QR code may be placed elsewhere on the scent dispenser 812 in other

embodiments.

[0142] FIG. **31** shows an alternate “vertical” scent dispenser/holder combination **910**, which includes a cradle **912** and a stand **914**, which can be permanently fixed together in one piece, or removably coupled from one another. Cradle **912** includes an arcuate slot **916** adapted to receive a replaceable scent cartridge **918**, which can be selectively removed and replaced as needed. Scent cartridge **918** can be the same as one or more of the scent cartridges described hereinabove in accordance with various embodiments of the present invention. Cradle **912** includes a front face **920** having front openings **922** and a rear face **924** having rear openings **926**. Compared to the embodiment of FIGS. **28-30**, which has a uni-directional configuration for emitting scents, scent dispenser/holder combination **910** has a “multi-directional” scent emitting configuration. For instance, scent can exit through front openings **922**, rear openings **926** and/or arcuate slot **916**.

[0143] In the embodiment depicted in FIG. **31**, the width of arcuate slot **916** is selected to accommodate a single scent cartridge **918**. However, in alternate embodiments, this width can be adjusted, for instance, to accommodate more than one scent cartridge. If all cartridges contain the same scent, the strength of that scent may be increased depending upon the number of cartridges employed. In an alternate embodiment, two or more cartridges containing different scents could be loaded into cradle **912**. For instance, a first scent could be diffused through front openings **922**, a second scent could be diffused through rear openings **926**, and a mixture of the first and second scents could be diffused through arcuate slot **916**. In other embodiments, front face **920** or rear face **924** could be void of its associated openings (i.e., front openings **922** and rear openings **926**, respectively).

Examples: Experimental Validation

[0144] FIG. **24** is a graphical representation of diffusion curves comparing the scent insert of the present invention (solid line) with a commercial diffusion-type air freshener. For comparison, the scent diffusion curve of the present invention has been multiplied by a factor of three, while the compared product had an initial mass eleven times larger than that of the scent insert made in accordance with an embodiment of the present invention. In this test, the inventive insert had a ratio of 1.85 grams of scent per gram of the insert. At the conclusion of this test, the ratio was 1.65. The samples were carrying equivalent scents.

[0145] FIG. **25** is a graphical representation of diffusion curves comparing the scent insert of the present invention (solid line) with a commercial diffusion-type air freshener for a different scent. For comparison, the scent diffusion curve of the present invention has been multiplied by a factor of eight, while the compared product sample had an initial mass fifteen times larger than that of the scent insert made in accordance with an embodiment of the present invention. In this test, the inventive insert had a ratio of 0.78 grams of scent per gram of the insert. At the conclusion of this test, the ratio was 0.72. The samples were carrying equivalent scents.

[0146] FIG. **26** illustrates the performances of various hydrophilic scented materials. This chart offers a compilation of data showing the impacts of pore size and density of the hydrophilic core material when used with various odorants as well as water and alcohol. No enclosing membrane was present during these tests. The circles labeled NP1 are constructed with a typical known process. The circles labeled BSHC were a thinner material with greater pores per inch and higher density of hydrophilic material compared to the NP1 material. The circles labeled BS were a thinner material with greater pores per inch but having the typical density of the NP1 hydrophilic material. The circles labeled HC were the typical material NP1 but with higher density hydrophilic material.

[0147] FIG. **27** shows the effect of selecting various membrane materials for the release of scent over time. The tests were conducted with a commercial oil formulation as the scent. It should be noted that this formulation was not optimized for diffusion in the tested materials, foam or membranes. A cross all categories, relatively linear diffusion characteristics were seen for approximately the first 40% of scent emitted. The data demonstrate that high vinyl acetate

concentrations in EVA and/or thinner layers increase diffusion rate and reduce active life. In contrast, lower vinyl acetate concentrations in EVA and/or thicker layers decrease diffusion rate, while increasing active life. In this manner, the scent dispensing parameters can be modulated depending on the intended application.

[0148] Many, if not all, of the scent-dispensing devices disclosed herein can be adapted for use in connection with smell retraining therapy, which typically involves the utilization of four specific scents, namely, “clove,” “rose,” “lemon,” and “eucalyptus.” The clinically recommended protocol calls for a patient to smell each of these four scents for fifteen seconds twice a day for several weeks to several months. While many, if not all, of the scent-dispensing devices disclosed herein can be adapted for use as a smell (i.e., olfactory) retraining aid, the scent dispensers disclosed in aforementioned U.S. Pat. No. 8,544,766 are especially adapted for such use by providing them in the form of a kit containing one or more sets of four scent dispensers. Each set would include a scent dispenser containing a corresponding one of the scents recommended for smell retraining therapy. In other words, and by way of example, such a kit would include: (i) a scent dispenser provided with an essential oil having a “clove” scent, (ii) a scent dispenser provided with an essential oil having a “rose” scent, (iii) a scent dispenser provided with an essential oil having a “lemon” scent, and (iv) a scent dispenser provided with an essential oil having a “eucalyptus” scent. A user following the aforementioned smell retraining protocol could then diffuse these four scents in a dry air formulation without any appreciable adulteration of the diffused scent.

[0149] Reference will now be made to the exemplary embodiments of the present invention illustrated in FIGS. 32-37 and described in connection therewith. Wherever practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict the exemplary embodiments of the present invention for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated in FIGS. 32-37 may be employed, without departing from the objects of the invention described hereinabove.

[0150] Referring now to FIG. 32, an exploded view of a scent dispenser/absorber **1010** is shown. Specifically, the parts illustrated in FIG. 32 are ordered to show their orientation when scent dispenser/absorber **1010** is in its “on” condition. Accordingly, as shown in FIG. 33, the scent dispenser/absorber **1010** includes, as viewed from left to right, a vented cage **1012**, a scent cartridge **1014**, a base **1016**, and a lid **1018**.

[0151] When lid **1018** is removed from base **1016**, which includes finger grips **1020** to facilitate such removal, scent cartridge **1014** is exposed to the environment via a plurality of vents **1022** in cage **1012**. Thus, when the scent dispenser/absorber **1010** is in use in its “on” condition, the scent cartridge **1014** communicates with the outside environment via the vents **1022** in the cage **1012**. In a scent-dispensing mode, the vents **1022** allow scent to migrate from the scent cartridge **1014** to the surrounding environment. Conversely, in a scent-absorbing mode, the vents **1022** allow scent to migrate from the surrounding environment to the scent cartridge **1014**. For reasons to be detailed later, cage **1012** can include internal threads **1024**, as well as a plurality of finger grips **1026**.

[0152] Scent cartridge **1014** is adapted to be receivable in an open end **1028** of base **1016**, opposite a closed end **1030** of base **1016**. The base **1016** also includes an annular sidewall **1032** having external threads **1034**, for purposes to be described hereinbelow. In an embodiment, closed end **1030** of the base **1016** contains a raised annular rim **1036**, which includes external threads **1038** adapted to threadably mate with the internal threads **1024** of the cage **1012**. This can be via a “screw-on” and “screw off” mechanism of attachment, in which case, finger grips **1026** would aid the user in attaching and detaching cage **1012** from base **1016**. It should be understood that any alternate means for attaching/detaching cage **1012** from base **1016** could be employed, including snap-fit and bayonet mechanisms. Annular rim **1036** defines a compartment **1040** so that at least one scent cartridge **1014** can be locked into compartment **1040** via attachment of cage **1012** to base **1016**.

[0153] It is through the disposition of lid **1018** that the scent dispenser/absorber **1010** can be changed from its off position to its on position and vice-versa. More particularly, the lid **1018** has internal threads **1042** adapted to threadably mate with an adjacent set of the external threads **1034** located proximal the open end **1028** of the base **1016**, thereby removably joining the lid **1018** to the open end **1028** of the base **1016**. This form of attachment would be via a “screw on” and “screw off” mechanism. To facilitate such attachment and detachment, lid **1018** can include ridges/knurls **1044** to improve a user's grip. However, the same form of joinder may be accomplished via any number of known joining methods, such as dimples, snap fits, press fits, notches, friction fits, pins, magnets, etc. Lid **1018** and/or base **1016** can be made from glass, polypropylene or related materials with similar properties.

[0154] Through removal of the lid **1018** from the open end **1028** of the base **1016**, the scent-dispenser/absorber **1010** is put into its “on” position. However, for convenience, lid **1018** can be screwed onto closed end **1030** of base **1016** by mating its internal threads **1042** with an adjacent set of the external threads **1034** located proximal the closed end **1030** of the base **1016**, thus providing a convenient disposition of lid **1018** while maintaining scent dispenser/absorber **1010** in its “on” condition.

[0155] As indicated above, the cage **1012** operates to secure scent cartridge **1014** within compartment **1040** of base **1016**. In some embodiments, the cage **1012** and/or base **1016** can be sized and shaped to hold multiple scent cartridges holding the same scent, or different scents. For the purpose of preventing intermingling of scents, especially in the case of cartridges holding different scents, a thin, disc-like divider (not shown) can be removably placed between adjacent scent cartridges. Such a divider would ideally be impermeable to scent molecules. Removal of the divider can shift the scent dispenser/absorber **1010** from a single-scent mode to a composite mode wherein different scents are intermingled and dispensed as a customized scent. In another embodiment, the relative positioning of the cartridges can be switched to select which scent cartridge is being diffused into the environment (i.e., atmosphere). In a further embodiment, one of the scent cartridges can be adapted for a scent-diffusing mode, while another scent cartridge is adapted for a scent-absorbing mode. In such a configuration, the relative positioning of the scent cartridges can be chosen to switch the scent dispenser/absorber **1010** from a scent-diffusing mode to a scent-absorbing mode and vice-versa.

[0156] In one embodiment, the scent cartridge **1014** comprises reticulated hydrophobic polyurethane foam (Product No. RT030CHRSC1) from Ionac (Woodbridge INOAC Technical Products, 100 Carol Place, Moonachie, NJ 07074), which material provides the scaffolding for the scent cartridge **1014**. It is a desirable material for the scaffolding of the scent cartridge **1014** because scent molecules only weakly interact with it. Due to the fact that polyurethane molecules are not volatile at typical pressures and temperatures, the odorant scent released from the scent cartridge **1014** will not be adulterated by the scaffolding (i.e., the aforementioned polyurethane foam).

[0157] To prepare the scent cartridge **1014**, the foam or scaffolding material can be coated with a thin layer of prepolymer, such as Dow Chemical's HYPOL JM 5005, which can then be reacted with water molecules to form a hydrophilic polyurethane surface coating. The large hydrophilic polyurethane molecules are non-volatile and do not contribute to the scents produced. The hydrophilic polyurethane is polar in nature and has a degree of affinity for scent molecules.

[0158] As the hydrophilic polyurethane surface coating is curing, CO.sub.2 out gassing results in a larger surface area. On the surface of the reticulated material, the scent molecules adhere along the hydrophilic polyurethane surface due to their polar interactions. There is a large surface area available for evaporation and a “headspace” within the material that generally contains a high level of “dry air” scented vapor. Through this process, the scent cartridge **1014** can release a high-quality representation of the original scent, while ensuring that the scent lasts a long time. Various thicknesses (e.g., 3/8”) and pore sizes (e.g., 30 pores per inch) of reticulated base material are

possible.

[0159] The result is that when the scent dispenser/absorber **1010** is switched to its “on” condition, as described above, a faithful iteration of the original odorant scent will be released for sampling or other purposes. In other words, the aforementioned sampling step is carried out in an environment in which essentially ambient air contains scent molecules, but no liquid. Thus, the scent dispenser/absorber **1010** allows scents to be stored, transported and sampled in a non-liquid form, the sampling being performed in a “passive” manner.

[0160] While the scent cartridge **1014** has been described with reference to a particular embodiment thereof, it should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present invention. For instance, the scent dispenser/absorber **1010** is adapted for use with other types of scent capturing and diffusing media such as the scent cell disclosed in U.S. Patent Application Publication No. 2017/0312380 (see especially Paragraphs to [0050]), U.S. Pat. No. 8,544,766 (see especially Column 6, lines 30-59) and U.S. Patent. No. 6,617,014, all of which patent publications are incorporated herein by reference in their entireties.

[0161] In addition, many modifications may be made to the scent cartridge **1014** described herein to adapt it to a particular situation, use or application without departing from the overall objective, spirit and/or scope of the present invention. For instance, the scent cartridge **1014** could be replaced by a scent cartridge adapted to absorb and/or adsorb a volatile scented substance, while allowing the ready passage of air therethrough. In some other embodiments, the scent cartridge **1014** could contain an amount of volatile scented substance, such as those used in perfumes, for attracting game, for calming a pet, as diet aids, for aroma therapy, for medical applications, or for other uses which are known or may become known, such as providing cannabis-derived and/or hemp-derived scents, fragrances, etc. In some embodiments, the scent cartridge **1014** would be designed such that the scented substance may be added directly to the scent cartridge **1014** to replenish or change the scent.

[0162] By way of further example, the scent cartridge **1014** may be made of any material that can carry and release volatile scented substances. In some embodiments, the scent cartridge **1014** could be made of an absorbent fibrous material or closed cell foam having air passages penetrating therethrough. In other embodiments, the scent cartridge **1014** could be made of an open-cell foam that presents an appreciable ratio of surface area to volume of foam, with higher ratios typically being preferred. In such embodiments, the foam may be a hydrophilic foam or have a hydrophilic material exposed at the surfaces of its cells. Other suitable embodiments could utilize an open-cell foam composite made of substantially hydrophobic foam to provide structure to the composite and substantially hydrophilic foam exposed at the surfaces of its cells, such as the foam described in U.S. Pat. No. 6,617,014, whose disclosure is incorporated herein by reference in its entirety.

[0163] In yet other embodiments, the scent cartridge **1014** may comprise a nonwoven fibrous material substrate coated with, for example, a substantially hydrophilic foam coating which is exposed at the surface and in interstitial spaces within the nonwoven fibrous material. The interstitial spaces within the nonwoven fibrous material form air passages penetrating therethrough to allow the flow of air. Examples of suitable nonwoven fibrous materials include, without limitation, cotton, felt, silk, or combinations thereof. As will be recognized by persons of ordinary skill in the relevant art, such embodiments would be useful when the volatile scented substances employed to impart scent or alternative odor to the scent cartridge **1014** are of the types that may react with and degrade some hydrophobic foams (see, e.g., U.S. Pat. No. 8,544,766, which is incorporated herein by reference in its entirety). One possible process for producing such a scent cartridge involves contacting a substrate of nonwoven fibrous material with a prepolymer emulsion and then polymerizing or curing the emulsion. By way of further example, the substrate can be dipped or immersed in the prepolymer emulsion, which can also be applied by brushing or otherwise coating onto the substrate. In an embodiment of such a process, the substrate of

nonwoven fibrous material may be provided as a sheet or block and then coated with the prepolymer emulsion, followed by polymerization or curing of the emulsion to form the substantially hydrophilic foam on the nonwoven fibrous substrate. The substrate can then be cut into appropriately sized and shaped pieces to produce the desired scent cartridge.

[0164] In other embodiments, the scent cartridge **1014** could take the form of a square patch (e.g., 1" by 1"), or any other convenient shape. As a patch, it could be covered by fabric or be bare. In certain embodiments, it could be carried in a pocket of a user's clothes, or can even be integrated into the garments themselves, whether the clothes are for a human or an animal. A main objective of such embodiments is to provide a scented element that does not wick onto the clothes themselves. By way of example, insect-repellant scents in clothes could be useful for gardeners, while anti-seasickness scents in clothes could be useful for sailors and the like. To counteract foul odors, garbage cans and/or diaper bags could be integrated with odor-combatting scents.

[0165] The scent cartridge **1014** can be constructed in accordance with still further embodiments thereof. For instance, another alternate scent cartridge arrangement is disclosed in U.S. Provisional Patent Application Ser. No. 63/028,433, which is incorporated herein by reference.

[0166] In an embodiment, the replaceable and interchangeable scent cartridges could be provided in combination with replacement cages that would be essentially identical to cage **1012**. In other embodiments, the interchangeable and replaceable scent cartridges would not be associated with a replacement cage and would instead be adapted to be swapped into the cage **1012** of a user's previously purchased scent dispenser/absorber **1010**. In one embodiment, this can be done by removing the cage **1012**, removing the previous scent cartridge **1014** or replaceable and interchangeable scent cartridge (if any), optionally disinfecting the cage **1012**, inserting a new interchangeable and replaceable scent cartridge and reattaching the cage **1012** to the base **1016**.

[0167] For convenience of use, the lid **1018** includes an attachment clip **1046**, which can be removable, such as via magnets on clip **1046** and/or on base **1016**. This clip **1046** can allow for scent dispenser/absorber **1010** to be wearable, via attachment to garment(s), and/or to be removably attachable to sun visors on any type of vehicles (e.g., cars, boats and planes). In embodiments in which base **1016** is magnetized, the scent dispenser/absorber **1010** can also be removably coupled to a metal surface, such as on a refrigerator or in a car.

[0168] In yet another embodiment of the present invention, the inventive scent dispenser could be combined with certain scent-dispensing embodiments disclosed hereinabove. Specifically, scent dispenser/absorber **1010** of the present application could be removably coupled or permanently fixed within skirt-like ring **28** of FIGS. **1-4**. In such a combination, scent dispenser/absorber **1010** could operate as described hereinabove when the scent dispenser of FIG. **1** is in a closed state, by selectively removing/replacing lid **1018**. At the same time, a second scent dispensing functionality could be achieved by inverting cap **27** to activate a second scent-dispensing mode, with the same scent or a different scent as that provided by scent dispenser/absorber **1010**. In the first instance (i.e., closed state), the scent cartridge **20** is housed in internal chamber **26** and scent dispenser/absorber **1010** of the present invention is external. In the second configuration (i.e., second scent dispensing functionality), scent cartridge **20** is external and scent dispenser/absorber **1010** of the present invention is housed inside internal chamber **26**.

[0169] As illustrated in FIG. **35**, certain embodiments of the scent dispenser/absorber **1010** include a QR code Q (i.e., on the closed end **1030** of the base **1016**). The QR code Q is configured for the purpose of providing access to instructions, advertising, and/or other information. While the QR code is shown as being placed on the closed end **1030** of the base **1016**, the QR code may be placed elsewhere on the scent dispenser/absorber **1010** in other embodiments.

[0170] Reference will now be made to the exemplary embodiments of the present invention illustrated in FIGS. **38-42** and described in connection therewith. Wherever practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict the exemplary embodiments of the present invention for purposes of illustration

only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated in FIGS. **38-42** may be employed, without departing from the objects of the invention described hereinabove.

[0171] With the foregoing prefatory comments regarding FIGS. **38-42** in mind, reference is now made to FIG. **38**, which is an exploded view of a disposable scent dispenser/absorber **1210**. Specifically, the parts illustrated in FIG. **38** are ordered to show their orientation when scent dispenser/absorber **1210** is in its “on” condition (as also shown in FIGS. **41-42**). The scent dispenser/absorber **1210** includes a vented cage **1212**, a non-replaceable scent-diffusing cartridge **1214**, a thermoplastic membrane, or film, **1216**, and a base **1218**. As shown in FIGS. **39** and **41** and further discussed below, the vented cage **1212** and base **1218** are joined together to form a housing **1220**. The vented cage **1212** includes a perforated end **1221** that defines a plurality of vents **1222**, as further discussed below, and an open end **1223** opposite the perforated end **1221**.

[0172] As disclosed in connection with other embodiments discussed herein, the scent-diffusing cartridge **1214** is adapted for non-active, dry air scent diffusion in which scent molecules can be passively emitted in non-droplet form. The scent-diffusing cartridge **1214** is also adapted to be receivable in an open end **1225** of the base **1218**, opposite a closed end **1226** of the base **1218**.

[0173] The thermoplastic film **1216** is adapted to be receivable in the open end **1223** of the vented cage **1212**. The thermoplastic film **1216** is fixedly attached to the perforated end **1221** of the vented cage **1212**, preferably with an adhesive.

[0174] Referring again to FIGS. **39** and **41**, the vented cage **1212** and base **1218** are united to form the housing **1220** and enclose the scent-diffusing cartridge **1214** and thermoplastic film **1216** therein (i.e., within an interior chamber of the housing **1220**). To form the housing **1220**, the open end **1223** of the vented cage **1212** and the open end **1224** of the base **1218** are permanently (i.e., non-removably) attached to each other (i.e., at a fusion point or joint **1228** that extends around the circumference of the housing **1220**). The housing **1220** is configured to permanently contain the scent-diffusing cartridge **1214** and thermoplastic film **1216** therein. It should be understood that any means for attaching the vented cage **1212** to the base **1218** may be employed, including, but not limited to a threaded/“screw on” mechanism, dimples, snap fits, press fits, notches, friction fits, pins, magnets, sealants and other adhesives. Given the permanent attachment of the vented cage **1212** to the base **1218**, both the scent-diffusing cartridge **1214** and thermoplastic film **1216** are non-replaceable.

[0175] The vented cage **1212** and/or base **1218** can be made from glass, polypropylene or related materials with similar properties. To increase enjoyment, the vented cage **1212** and/or base **1218** can have a variety of amusing colors, shapes and designs.

[0176] As disclosed in connection with other embodiments discussed herein, the scent-diffusing cartridge **1214** comprises reticulated hydrophobic polyurethane foam (Product No. RT030CHRSC1) from Ionac (Woodbridge INOAC Technical Products, 100 Carol Place, Moonachie, NJ 07074) in various embodiments of the scent dispenser/absorber **1210**. The reticulated hydrophobic polyurethane foam provides the scaffolding for the scent cartridge **1214**. It is a desirable material for the scaffolding of the scent-diffusing cartridge **1214** because scent molecules only weakly interact with it. Due to the fact that polyurethane molecules are not volatile at typical pressures and temperatures, the odorant scent released from the scent-diffusing cartridge **1214** will not be adulterated by the scaffolding (i.e., the aforementioned polyurethane foam). In certain embodiments, the scent-diffusing cartridge **1214** may also comprise the NP1 hydrophilic material discussed hereinabove.

[0177] While the scent-diffusing cartridge **1214** has been described with reference to a particular embodiment thereof, it should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present invention. For instance, the scent dispenser/absorber **1210** is adapted for use with other types of scent capturing and diffusing media such as the scent cell disclosed in U.S. Patent

Application Publication No. 2017/0312380 (see especially Paragraphs [0048] to [0050]), U.S. Pat. No. 8,544,766 (see especially Column 6, lines 30-59) and U.S. Patent. No. 6,617,014, all of which patent publications are incorporated herein by reference in their entireties.

[0178] As disclosed in connection with other embodiments discussed herein, the thermoplastic film **1216** may comprise one or more polymers selected from the group consisting of HDPE; LDPE; PP and EVA in various embodiments of the scent dispenser/absorber **1210**. When the polymer is EVA, the EVA can be selected such that the EVA has (i) a VA concentration in a range of from about 3.5% to about 18.5% of the EVA, and (ii) a thickness in a range of from about 2 millimeters to about 6 millimeters. When the polymer is HDPE, the HDPE can have a thickness in a range of from about 1 millimeter to about 6 millimeters. When the polymer is LDPE, the LDPE can have a thickness in a range of from about 1 millimeter to about 6 millimeters.

[0179] As disclosed in connection with other embodiments discussed herein, the thermoplastic film **1216** performs a metering function when used in combination with the material of the scent-diffusing cartridge **1214** in that it prevents, or at least inhibits, the release of liquid, while permitting the release of scent molecules, thereby delivering scent to a user in a dry format free from mists, sprays, aerosols and liquid. In the embodiment illustrated in FIGS. **38-42**, the scent-diffusing cartridge **1214** and thermoplastic film **1216** do not directly contact one another. However, in other embodiments (not illustrated herein), the scent-diffusing cartridge **1214** and thermoplastic film **1216** are arranged within the interior chamber of the housing **1220** to be in direct contact with each other.

[0180] To facilitate their combination, the scent-diffusing cartridge **1214** and thermoplastic film **1216** can be shaped to complement one another. In the embodiment illustrated in FIGS. **38-42**, the scent cartridge **1214** and thermoplastic film **1216** both have a disc shape. However, it will be understood that the scent-diffusing cartridge **1214** and thermoplastic film **1216** may alternatively be configured to have any other desired shapes and dimensions.

[0181] Referring now to FIGS. **39** and **40**, the scent dispenser/absorber **1210** is shown in a “off” condition, or first position, in which the closed end **1226** of the base **1218** faces upward and the perforated end **1221** of the vented cage **1212** faces downward so as to be proximate to a horizontal support surface H. In this condition/position, the scent-diffusing cartridge **1214** is isolated from the exterior environment, whereby no scent is diffused from the housing **1220**.

[0182] As discussed above, and shown in FIGS. **41** and **42**, the scent-diffusing cartridge **1214** is exposed to the environment via a plurality of vents **1222** formed in the perforated end **1221** of the cage **1212**. When the scent dispenser/absorber **1210** is in use in its “on” condition, or a second position, the perforated end **1221** of the vented cage **1212** faces upward and the closed end **1226** of the base **1218** faces downward so as to be in contact with the horizontal support surface H, as shown in FIG. **41**. When in the “on” condition/second position, the scent-diffusing cartridge **1214** is exposed to the exterior environment and able to communicate with the exterior environment via the vents **1222** in the cage **1212**.

[0183] In a scent-dispensing mode, the vents **1222** allow scent to migrate from the scent-diffusing cartridge **1214** to the surrounding environment. Conversely, in a scent-absorbing mode, the vents **1222** allow scent to migrate from the surrounding environment to the scent-diffusing cartridge **1214**.

[0184] Referring again to FIGS. **39** and **40**, when the scent dispenser/absorber **1210** is in its “off” condition, the perforated end **1221** of the vented cage **1212** directly contacts the horizontal support surface H, which covers the vents **1222** and thereby substantially prevents scent from migrating (i.e., via a gaseous transfer) from the scent-diffusing cartridge **1214** and the housing **1220** into the surrounding environment. In some embodiments, the scent migrates out of the scent dispenser/absorber **1210** and contacts the horizontal support surface H, but the actual amount of odorant transferred to the horizontal support surface H is minimal/very low (i.e., to the extent that no scent is detectable).

[0185] In other embodiments (not illustrated), the perforated end **1221** of the vented cage **1212** includes a gasket about its circumference, such that when the scent dispenser/absorber **1210** is in its “off” condition, the gasket directly contacts the horizontal support surface H and thereby defines a space within which any scent migrating from the scent-diffusing cartridge **1214** through the vents **1222** of the vented cage **1212** is contained and prevented (or substantially contained/prevented) from migrating to the surrounding environment.

[0186] To change the scent dispenser/absorber **1210** from its second position (i.e., its “on” condition) to its first position (i.e., its “off” condition), a user simply lifts the scent dispenser/absorber **1210** off of the horizontal support surface H and inverts, or “flips” the scent dispenser/absorber **1210** over, and places it on the horizontal support surface H in its “off” condition/first position (as shown in FIGS. **39** and **40**). The user completes the same inverting/flipping action to change the scent dispenser/absorber **1210** from its “off” condition back to its “on” condition/second position (as shown in FIGS. **41** and **42**).

[0187] As illustrated in FIGS. **38** and **40**, certain embodiments of the scent dispenser/absorber **1210** include a QR code Q (i.e., on the closed end **1226** of the base **1218**). The QR code Q is configured for the purpose of providing access to instructions, advertising, and/or other information. While the QR code is shown as being placed on the closed end **1226** of the base **1218**, the QR code may be placed elsewhere on the scent dispenser/absorber **1210** in other embodiments.

[0188] It will be understood that the embodiments described hereinabove are merely exemplary and that a person skilled in the art may make many variations and modifications without departing from the spirit and scope of the present invention. For instance, besides the “passive” embodiments described above, “active” (i.e., driven) embodiments of the present invention are also possible. Furthermore, the present invention is not necessarily limited to “dry air” embodiments. All such variations and modifications are intended to be included within the scope of the present invention.

Claims

1. A method for attracting game, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful for attracting game; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent is dischargeable from said compartment to an outside environment containing game.
2. The method of claim 1, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.
3. A method for calming a pet animal, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful for animal husbandry; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent

is dischargeable from said compartment to an outside environment containing a pet animal.

4. The method of claim 3, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.

5. A method for aiding medical diagnostics of a patient, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful for medical diagnostics; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent is dischargeable from said compartment to an outside environment containing a patient.

6. The method of claim 5, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.

7. A method for facilitating aroma therapy of an individual, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful in aroma therapy; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent is dischargeable from said compartment to an outside environment containing an individual.

8. The method of claim 7, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.

9. A method for facilitating the dieting of an individual, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful as a diet aid; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent is dischargeable from said compartment to an outside environment containing a dieting individual.

10. The method of claim 9, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.

11. A method for facilitating insomnia therapy of an individual, comprising the steps of: providing an apparatus for dispensing scent, said apparatus including a housing having an open end, a closed end opposite said open end, and an interior chamber between said open end and said closed end, and said apparatus further including an invertible lid having an associated compartment configured to allow the discharge of scent therefrom, said lid being attachable to said open end of said housing in a first orientation, in which said compartment is positioned entirely within said interior chamber of said housing between said lid and said closed end of said housing, and in a second orientation, in

which said compartment is positioned externally of said housing; providing said compartment with a scented material configured to emit a pre-selected scent useful in insomnia therapy; and attaching said lid to said open end of said housing in said second orientation, whereby said pre-selected scent is dischargeable from said compartment to an outside environment containing an individual who is experiencing insomnia.

12. The method of claim 11, wherein said scented material is configured to emit said pre-selected scent in a dry air manner in which scent molecules are emitted in non-droplet form.

13. A combination, comprising: a reticulated foam composite having a capacity sufficient to receive a quantity of liquid odorant; and a thermoplastic membrane enveloping said reticulated foam composite.

14. The combination of claim 13, wherein said reticulated foam composite comprises a scaffold which includes a hydrophobic core and a hydrophilic coating applied to said hydrophobic core.

15. The combination of claim 14, wherein said thermoplastic membrane is heat-sealed around said scaffold.

16. The combination of claim 13, wherein said hydrophobic core of said scaffold is a first foam material and said hydrophilic coating of said scaffold is a second foam material.

17. The combination of claim 16, wherein said first foam material has an open cell structure, including a plurality of pores defined by a plurality of surfaces.

18. The combination of claim 17, wherein said thermoplastic membrane comprises one or more polymers selected from the group consisting of HDPE; LDPE; PP and EVA.

19. The combination of claim 18, wherein said thermoplastic membrane is an EVA polymer film.

20. The combination of claim 19, wherein said EVA polymer film is selected such that said EVA has a VA concentration in a range of from about 3.5% to about 18.5%.

21. The combination of claim 20, wherein said VA concentration is about 18.5% of said EVA.

22. The combination of claim 21, wherein said EVA polymer film has a thickness in a range of from about 2 millimeters to about 6 millimeters.

23. The combination of claim 20, wherein said VA concentration is about 3.5% of said EVA.

24. The combination of claim 23, wherein said EVA polymer film has a thickness in a range of from about 2 millimeters to about 6 millimeters.

25. The combination of claim 18, wherein said polymer is HDPE.

26. The combination of claim 25, wherein said HDPE has a thickness in a range of from about 1 millimeter to about 6 millimeters.

27. The combination of claim 18, wherein said polymer is LDPE.

28. The combination of claim 27, wherein said LDPE has a thickness in a range of from about 1 millimeter to about 6 millimeters.

29. The combination of claim 14, wherein said hydrophobic core of said scaffold has a volume in a range of from about 0.1 cubic inches to about 4 cubic inches.

30. The combination of claim 14, wherein said hydrophobic core of said scaffold has a weight in a range of from about 150 milligrams to about 6 grams.

31. The combination of claim 14, wherein said scaffold contains liquid odorant, whereby said combination is adapted to function as a scent-producing device.

32. The combination of claim 31, wherein said hydrophobic core of said scaffold absorbs said liquid odorant and releases scent molecules emanating therefrom, at least some of said scent molecules released from said hydrophobic core being discharged from said scaffold through said hydrophilic coating thereof.

33. The combination of claim 32, wherein said thermoplastic membrane has an inner surface bordering said hydrophilic coating of said scaffold and an outer surface opposite said inner surface, said inner surface having a first concentration of said scent molecules released from said hydrophobic core and said outer surface having a second concentration of said scent molecules discharged from said scaffold, said second concentration being less than said first concentration.

- 34.** The combination of claim 33, wherein said thermoplastic membrane is permeable to said scent molecules such that at least some of said scent molecules discharged from said scaffold are diffused through said thermoplastic membrane.
- 35.** The combination of claim 34, wherein said thermoplastic membrane is impermeable to liquid emanating from said liquid odorant.
- 36.** The combination of claim 35, wherein said at least some of said discharged scent molecules are diffused through said thermoplastic membrane as dry, but scented, air.
- 37.** The combination of claim 36, wherein said diffused scent molecules create a relatively low level of residual scent on said outer surface of said thermoplastic membrane, said level of residual scent being dependent upon the evaporation rate of said liquid odorant.
- 38.** The combination of claim 37, wherein said evaporation rate of said liquid odorant is in a range of from about 5 mg per day to about 25 mg per day.
- 39.** The combination of claim 38, wherein said hydrophobic core of said scaffold is provided with said liquid odorant in an amount in a range of from about 50 milligrams to about 12 grams.
- 40.** A disposable scent-diffusing apparatus, comprising: a permanently sealed housing having a perforated end, a closed end opposite said perforated end and an interior chamber between said closed end and said perforated end; a scent-diffusing cartridge positioned within said interior chamber adjacent said perforated end, said scent-diffusing cartridge being adapted for non-active, dry air scent diffusion in which scent molecules can be passively emitted in non-droplet form, said scent-diffusing cartridge being selectively isolatable from and exposed to an exterior environment surrounding said apparatus, wherein said scent-diffusing cartridge is isolated from the exterior environment when said apparatus is in a first position, and wherein said scent-diffusing cartridge is exposed to the exterior environment when said apparatus is in a second position; and a thermoplastic film provided within said interior chamber, between said perforated end of said housing and said scent-diffusing cartridge.
- 41.** The apparatus of claim 40, wherein said scent-diffusing cartridge comprises an NP1 hydrophilic material.
- 42.** The apparatus of claim 40, wherein said scent-diffusing cartridge comprises reticulated hydrophobic polyurethane foam.
- 43.** The apparatus of claim 40, wherein said thermoplastic film comprises one or more polymers selected from the group consisting of HDPE; LDPE; PP and EVA.
- 44.** The apparatus of claim 40, wherein said housing is made from a material selected from the group consisting of glass and polypropylene.
- 45.** The apparatus of claim 40, wherein said closed end includes a QR code on an exterior surface thereof.
- 46.** A n apparatus for diffusing scent into an external environment, comprising: a housing having a first end adapted for communication with the external environment, a second end opposite said first end, said second end being out of communication with the external environment, and an interior chamber between said first and second ends and surrounded by an exterior surface of said housing; a scent-diffusing cartridge positioned within said interior chamber of said housing; and a QR code placed on said exterior surface of said housing at a location where it is visible from the external environment.
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